

Computer Assisted Composition

Composition process - prerequisites

- Composition process as interaction of intuition and structure, "self-reflexive improvisation in non-real time".
- Reflects on compositional processes and gains experience in the relationship between musical conception and its compositional or acoustic reality
- Contemporary composition always arises at the intersection of technical and aesthetic prior knowledge and its speculative extrapolation.
The computer integrates these aspects into the creative process and overcomes the temporal lag between composition, rehearsal and performance found in traditional instrumental composition.

Computer Assisted Composition (CAC)

- Formalization leads to the further development of the compositional idea
 - Reflection on one's own musical thinking
- Evaluation of perceptual aspects of a musical idea through acoustic verification (e.g. speed, density, orchestration, etc.)
- Recursive analysis as an overcoming of the gap between compositional structure and acoustic result
 - Focus on model and process → Execution by the computer
 - Critical handling of own results through repeatability of the process with changed parameters
- Integration of extra-musical influences or data
 - Extension of control of musical parameters
- Abolition of the academic separation between instrumental and electronic music

Modeling options in Opusmodus

- Pitches
 - Frequency-based (spectral) pitch organization,
 - Transposition
 - Instrument ranges
- Durations
 - formal durations (number of events per process - density)
 - Entry points (diminution - augmentation)
 - rhythmic durations
- Dynamics
 - Independence of dynamics (as non-expressive parameter)
- Instruments
 - instrumentation
 - number of voices

Instrument-specific quantization of processes in Opusmodus

Notation (feasibility)

- Scaling of Complexity (Resolution)

Pitch:

- Instrument ranges
- Semi-, quarter-tones, etc.
- Intervals

Density:

- Speed (tempo vs. rhythmic notation)
- Rhythmic complexity (tuplets)
- Differentiation between entry distance and duration of events

Randomness / Probability in Opusmodus

- For every compositional decision, there is always more than one solution
- Discovery of the potentials through variants of a basic musical idea
- A musical model (genotype) → variety of possible variations (phenotype)
- Physical time runs as the most direct connection (straight line) between two points in time
- Music modulates physical time ("Erlebniszeit", Stockhausen)
- Resolution of linear i. foreseeable structures by controlled randomness while maintaining the general direction between two time points (linear, exponential, random)

